Deep Verification of the VSEPR-SIM Structural Energy Kernel (v2.7)

Architecture, Implementation, and
Deterministic Validation

Formation Engine Development Team

Version 2.7.1 — March 8, 2026

Verification status	485 / 485 PASS, 0 FAIL, 0 SKIP
Deep suite	256 / 256 PASS
Phase executables	229 / 229 PASS
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Tag	`v2.7.1`

MIT License | C++20 | CMake 3.15+

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1 Abstract

VSEPR-SIM is a classical atomistic formation engine that generates molecular and crystal structures deterministically from elemental composition and thermodynamic boundary conditions. This document presents the architecture, energy model, numerical methods, and verification framework of the v2.7.1 release.

The deep verification suite comprises 256 individual checks spanning Lennard-Jones parameter audits (42 UFF elements), Coulomb electrostatics, dielectric screening across 19 experimental solvents, crystal geometry validation against 30 Wyckoff reference lattices, molecular bond detection, composite force–energy consistency, noble-gas cluster FIRE relaxation with two-layer convergence criteria, restoring-force scans, crystal builder geometry fidelity, randomised seeded consistency suites, NVE energy conservation, and cross-checks against PubChem and NIST spectroscopic reference data.

All 256 deep verification checks and 229 phased revalidation checks pass without failure or skip. NVE energy drift remains below 3×10^{-11} kcal/mol on argon clusters. FIRE minimisation converges to $F_{\mathrm{rms}} < 10^{-5}$ kcal/mol/Å for all tested noble-gas clusters (Ar_2–Ar_7).

This result establishes the first fully verified deterministic kernel baseline for VSEPR-SIM, marking the transition from foundational correctness recovery toward convergence tightening, expanded physical coverage, and formation-oriented workflow integration.

2 Project Motivation

Atomistic simulation engines fail silently when:

- Force and energy terms diverge (inconsistent analytic gradient vs. numerical derivative)
- Geometric emitters produce incorrect crystal structures (wrong atom count, wrong nearest-neighbour distance)
- Electrostatic scaling breaks (dielectric screening not proportional to ε_r)
- Numerical drift accumulates (NVE total energy not conserved)
- Parameter tables are incomplete (fallback to default values without warning)

These failure modes are difficult to detect in production because individual simulations may produce plausible-looking results while carrying systematic errors. VSEPR-SIM addresses this through two principles:

1. **Deterministic kernel design.** Given identical inputs, the engine produces bit-identical outputs. All random number generators are seeded and logged.
2. **Verification-driven development.** Every physics module is exercised by a self-testing suite before integration into the production pipeline. New features must pass existing checks before being merged.

No fake physics—code reflects real physical modelling; visual-only approximations are never used.

3 Historical Development

Table 1: Release timeline.

Tag	Date	Milestone
v2.5.0	2025	Clean baseline release with 13-section methodology
v2.5.3	2025	Heat-gated reaction control system
v2.6.1	2025	Alpha model generative baseline
v2.6.2	2025	Plateau investigation and root-cause analysis
v2.6.3	2025	Empirical lookup table (118 elements, compile-time)
v2.7.0	2025	Qt6 desktop application scaffold
v2.7.1	**2025**	**Deep verification milestone (485/485)**

The progression from v2.5.0 to v2.7.1 reflects a deliberate arc: establish the methodology (v2.5), explore generative models (v2.6), diagnose and stabilise the kernel (v2.6.2–v2.6.3), and verify the result exhaustively (v2.7.1). The v2.6.2 "plateau investigation" was a critical turning point—it identified four root causes (f-block degeneracy, metric mismatch, coordinate-descent saddle points, and smoothness penalty) that motivated the shift from generative exploration to verification-first development.

4 System Architecture

4.1 Pipeline Overview

The formation engine processes structures through a layered pipeline:

Layer	Responsibility
1. Formation Priors	Elemental identity, stoichiometry, boundary conditions
2. Topology Generation	Bond detection, molecular graph, crystal lattice
3. Energy Kernel	LJ + Coulomb + bonded MM evaluation
4. Integration	FIRE minimisation, velocity Verlet, Langevin
5. Structure Output	XYZ / XYZA / XYZC with SHA-256 provenance
6. Verification	Automated self-testing against empirical references

4.2 Atom Representation

Each atom carries:

- Position $\mathbf{x}_i \in \mathbb{R}^3$ (Å)
- Velocity $\mathbf{v}_i \in \mathbb{R}^3$ (Å/fs)
- Force $\mathbf{F}_i \in \mathbb{R}^3$ (kcal/mol/Å)
- Mass m_i (amu)
- Atomic number Z_i

- Formal charge q_i (elementary charges)

The `State` object aggregates N atoms with parallel arrays for maximum cache locality. No atom stores pointers to other atoms—all interactions are computed from positions at evaluation time.

4.3 Model Interface

All force models implement a pure-function contract:

$$\texttt{eval} : \mathcal{S} \times \Theta \longrightarrow (\mathbf{F}, \mathcal{E}) \tag{1}$$

where $\mathcal{S}$ is the State, Θ is the parameter set (cutoff radius, force constants, environment context), $\mathbf{F}$ is the force array, and $\mathcal{E}$ is the six-component energy ledger (U_{LJ}, U_{Coul}, U_{bond}, U_{angle}, U_{torsion}, U_{external}).

This is a pure function: given the same State and parameters, it always returns the same forces and energies.

4.4 Desktop Application Layer

The v2.7.0 release introduced a Qt6 desktop application scaffold with a three-layer architecture:

Layer	Components
Presentation	MainWindow, ObjectTree, PropertiesPanel, ViewportWidget
Bridge	EngineAdapter (atomistic $\leftrightarrow$ Qt type conversion)
Engine	Atomistic kernel (unchanged from CLI pathway)

The bridge layer ensures the atomistic kernel remains independent of the GUI framework.

5 Energy Model

5.1 Total Energy

The total potential energy is:

$$U_{\mathrm{total}} = U_{\mathrm{LJ}} + U_{\mathrm{Coul}} + U_{\mathrm{bond}} + U_{\mathrm{angle}} + U_{\mathrm{torsion}} + U_{\mathrm{ext}} \tag{2}$$

Each term is independently computed and stored in the energy ledger.

5.2 Lennard-Jones 12-6 Potential

Pairwise van der Waals interactions:

$$U_{\mathrm{LJ}}(r_{ij}) = 4\varepsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r_{ij}}\right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}}\right)^{6}\right] \tag{3}$$

Parameters are drawn from the Universal Force Field (UFF) of Rappé et al. (1992). VSEPR-SIM v2.7.1 includes 42 elements with verified σ and ε values.

Lorentz–Berthelot mixing rules:

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2} \tag{4}$$

$$\varepsilon_{ij} = \sqrt{\varepsilon_i \cdot \varepsilon_j} \tag{5}$$

The equilibrium distance is:

$$r_{\min} = 2^{1/6}\sigma \tag{6}$$

5.3 Coulomb Electrostatics

Point-charge electrostatics with distance-dependent dielectric screening:

$$U_{\text{Coul}}(r_{ij}) = \frac{k_e\, q_i\, q_j}{\varepsilon_r\, r_{ij}} \tag{7}$$

where $k_e = 332.0637$ kcal/mol $\cdot$ Å$/e^2$ and ε_r is the relative dielectric constant.
The dielectric model supports:

- Near-vacuum ($\varepsilon_r = 1.0$) for gas-phase systems
- Explicit solvent dielectrics from CRC Handbook, 104th Edition (19 solvents from hexane $\varepsilon_r = 1.88$ to water $\varepsilon_r = 78.4$)

5.4 Bonded Terms

Bond stretch (harmonic):

$$U_{\text{bond}} = \frac{1}{2} k_r (r - r_0)^2 \tag{8}$$

Angle bend (harmonic):

$$U_{\text{angle}} = \frac{1}{2} k_\theta (\theta - \theta_0)^2 \tag{9}$$

Torsion (cosine series):

$$U_{\text{torsion}} = \sum_n \frac{V_n}{2} \left[1 + \cos(n\phi - \gamma_n)\right] \tag{10}$$

Force constants are derived from UFF where available.

5.5 Compile-Time Empirical Database

The `empirical_reference.hpp` header provides compile-time arrays:

Table	Entries
Bond lengths (experimental)	57
Bond angles	25
Crystal lattice parameters	30
LJ parameters (UFF)	42
Diatomic spectroscopic constants (NIST)	10
Solvent dielectrics (CRC 104)	19
Shannon ionic radii	15
Noble gas dimer parameters	5

All values carry tolerance bounds and literature source annotations.

6 Structural Systems

6.1 Bond Detection

Bonds are detected by covalent-radius overlap:

$$r_{ij} < f \cdot (R_{\text{cov},i} + R_{\text{cov},j}) \tag{11}$$

where $f = 1.15$ is the tolerance factor. The algorithm runs in $O(N^2)$ for small molecules and uses cell lists for larger systems.

Suite E validates this against 9 canonical molecules (H_2 through CH_4), confirming exact bond counts.

6.2 Crystal Emitter

Crystal structures are built from unit cell definitions with fractional coordinates, space group metadata, and lattice parameters:

Preset	**Type**	a **(Å)**	**Space group**
Cu	FCC	3.615	Fm$\bar{3}$m (225)
Fe	BCC	2.870	Im$\bar{3}$m (229)
NaCl	Rocksalt	5.640	Fm$\bar{3}$m (225)
Si	Diamond	5.431	Fd$\bar{3}$m (227)
Al	FCC	4.050	Fm$\bar{3}$m (225)
MgO	Rocksalt	4.212	Fm$\bar{3}$m (225)
CsCl	CsCl	4.123	Pm$\bar{3}$m (221)

Supercell replication is exact: an $n_a \times n_b \times n_c$ supercell contains $n_a \cdot n_b \cdot n_c \cdot Z_{\text{cell}}$ atoms, verified by Suite L for random (n_a, n_b, n_c) triplets.

6.3 Crystal Builder Geometry Fidelity

Suite Q verifies the builder output for four preset crystals:

- **Atom count**: $N = 8 \times Z_{\text{cell}}$ for all $2 \times 2 \times 2$ supercells
- **Nearest-neighbour distance**: within 3% of the Wyckoff reference value

Note: FIRE relaxation is *not* applied to crystal supercells in the current suite because UFF LJ $r_{\min}$ exceeds the actual bond length for metals (Cu, Fe) and covalent crystals (Si), placing atoms inside the LJ repulsive wall. For NaCl, the Coulomb–LJ equilibrium differs from the physical crystal geometry. This is documented as a known scope boundary.

7 Numerical Methods

7.1 FIRE Minimisation

The Fast Inertial Relaxation Engine (Bitzek et al., 2006) is the primary structure optimiser. The implementation uses Euler integration with a velocity kick after power-negative resets:

1. Evaluate forces $\mathbf{F}_i$ at current positions
2. Compute power $P = \sum_i \mathbf{v}_i \cdot \mathbf{F}_i$
3. Mix velocities: $\mathbf{v}_i \leftarrow (1-\alpha)\mathbf{v}_i + \alpha|\mathbf{v}|\hat{\mathbf{F}}_i$
4. If $P > 0$: increment positive-power counter; increase Δt and decrease α
5. If $P \leq 0$: kick $\mathbf{v}_i = (\Delta t/|\mathbf{F}|)\,\mathbf{F}_i$; decrease Δt; reset α
6. Update positions: $\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i \Delta t$

7.1.1 Euler–FIRE Deadlock Correction

A deadlock condition was identified and corrected during v2.7.1 development. In the original implementation, the $P < 0$ branch set $\mathbf{v}_i = \mathbf{0}$. Because the integrator is pure Euler ($\mathbf{x} \leftarrow \mathbf{x} + \mathbf{v}\Delta t$, no acceleration term), zero velocity means zero displacement, which yields $\Delta U = 0$ on the next evaluation, falsely triggering the ε_U convergence criterion.

The correction replaces the zero-velocity reset with a kick along the normalised force direction:

$$\mathbf{v}_i \leftarrow \frac{\Delta t}{|\mathbf{F}|}\,\mathbf{F}_i \qquad \text{(after } P < 0 \text{ event)} \tag{12}$$

and conditions the ε_U check on $\varepsilon_U > 0$.

7.2 Velocity Verlet (NVE)

The standard four-step symplectic integrator:

$$\mathbf{v}_i(t + \tfrac{\Delta t}{2}) = \mathbf{v}_i(t) + \frac{\mathbf{F}_i(t)}{m_i} \cdot \text{ACC_CONV} \cdot \frac{\Delta t}{2} \tag{13}$$

$$\mathbf{x}_i(t + \Delta t) = \mathbf{x}_i(t) + \mathbf{v}_i(t + \tfrac{\Delta t}{2}) \cdot \Delta t \tag{14}$$

$$\mathbf{F}_i(t + \Delta t) = -\nabla_{\mathbf{x}_i} U\big(\mathbf{X}(t + \Delta t)\big) \tag{15}$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t + \tfrac{\Delta t}{2}) + \frac{\mathbf{F}_i(t + \Delta t)}{m_i} \cdot \text{ACC_CONV} \cdot \frac{\Delta t}{2} \tag{16}$$

Suite M verifies energy conservation to $< 3 \times 10^{-11}$ kcal/mol on Ar_3–Ar_5 clusters.

7.3 Langevin Thermostat (NVT)

BAOAB splitting for the Langevin equation, providing canonical (NVT) sampling with a tunable friction coefficient γ.

8 Verification Framework

8.1 Design Philosophy

The verification framework is built on five principles:

1. **Self-contained executables.** Each phase is a standalone C++ program that prints PASS/-FAIL. No external test frameworks.

2. **Empirical grounding.** Reference values come from published literature (Rappé, Wyckoff, CRC, NIST) with explicit tolerances.
3. **Reproducibility.** Random suites are seeded; the seed is printed in every report. Re-running with `-seed N` reproduces the exact result.
4. **Layered severity.** Suite G demonstrates two-layer checks: Layer 1 (descent) is always required; Layer 2 (convergence quality) adds F_{rms}, overlap, and stagnation guards.
5. **Tiered workflow.** Quick local runs skip network-dependent Suite P; full release verification includes it.

8.2 Phased Revalidation (Phases 1–14)

Phase	Checks	Status
1 (kernel)	11	PASS
2 (structural)	15	PASS
3 (relaxation)	17	PASS
4 (presets)	52	PASS
5 (crystals)	13	PASS
6/7 (sessions)	43	PASS
8–10 (env)	28	PASS
11–13 (comp)	26	PASS
14 (milestones)	24	PASS
Total	**229**	**ALL PASS**

9 Deep Verification Suite

The deep verification suite (`apps/deep_verification.cpp`, 1213 lines) exercises the kernel across 16 suites.

9.1 Suite Catalogue

Suite	Description	Checks	Type
A	LJ parameter audit (42 UFF elements vs Rappé)	42	Static
B	Homonuclear dimer sweeps (energy, r_{min}, smoothness)	20	Static
C	Ion-pair Coulomb + force consistency	15	Static
D	Crystal geometry vs Wyckoff (30 structures)	90	Static
E	Molecule bond detection (9 molecules)	9	Static
F	Composite force–energy (H_2O, NH_3, CH_4, Ar_2)	4	Static
G	Noble gas cluster FIRE (Ar_2–Ar_7, two-layer)	30	Static
H	Dielectric screening (19 CRC 104 solvents)	19	Static
N	Restoring-force scan (He_2–Xe_2)	10	Static
Q	Crystal builder fidelity ($2 \times 2 \times 2$ supercell)	8	Static
I	Random LJ force–energy (N_{rand} pairs)	1	Randomised

Suite	Description	Checks	Type
J	Random Coulomb $1/r$ law (N_{rand} pairs)	2	Randomised
K	Random dielectric ratio (N_{rand} samples)	1	Randomised
L	Random supercell atom count (6 crystals)	6	Randomised
M	NVE energy conservation (Ar_3–Ar_5)	6	Randomised
P	Downloaded empirical refs (PubChem + NIST)	53	External
	Total	**256**	

9.2 Suite A: LJ Parameter Audit

For each of 42 UFF elements, verify:

- $\sigma > 0$
- $\varepsilon > 0$
- $r_{\min} = 2^{1/6}\sigma$ (analytic identity)

9.3 Suite B: Homonuclear Dimer Sweeps

For noble gases and selected elements, sweep $r \in [0.8\sigma, 3\sigma]$ and verify:

- Energy minimum occurs at $r = r_{\min}$ (within tolerance)
- Well depth equals $-\varepsilon$
- Energy curve is smooth (no discontinuities from switching)

9.4 Suite D: Crystal Geometry

For 30 reference crystals, build the unit cell and verify:

- Nearest-neighbour distance matches Wyckoff within tolerance
- Coordination number is correct
- Atoms per cell equals Z_{cell}

9.5 Suite G: Noble Gas Cluster FIRE (Two-Layer)

Starting geometry: Fibonacci sphere with radius $R = r_{\min}\sqrt{N}/3.0$, placing all pairs at $> r_{\min}$ (attractive well).

Layer 1 (descent):

- Energy is finite
- $U_{\text{final}} < U_0$ (energy decreased)

Layer 2 (convergence quality):

- $F_{\text{rms}} < 0.10$ kcal/mol/Å
- $r_{\text{nn}} > 0.55 \cdot r_{\min}$ (no atom overlap)
- $\Delta U/N < 10^{-4}$ (not stagnating)

9.6 Suite N: Restoring-Force Scan

For each noble gas (He, Ne, Ar, Kr, Xe), place a homonuclear dimer at $r = r_{\min} + \delta$ for $\delta \in \{-1.5, -1.0, -0.6, -0.3, +0.3, +0.6, +1.0, +1.5\}$ Å and verify:

- Force direction is restoring: $F_x \cdot \delta > 0$
- $r_{\min}$ is the energy minimum: $U(r_{\min} + \delta) > U(r_{\min})$ for all $\delta \neq 0$

9.7 Suite P: Downloaded Empirical References

The `scripts/fetch_empirical.py` downloader retrieves:

- 25 bond lengths from PubChem 3D conformers
- 20 diatomic spectroscopic r_e from NIST WebBook
- 15 solvent dielectrics from CRC Handbook, 104th Edition

Suite P cross-checks these against the hardcoded database and runs solvent-specific Coulomb screening tests. The downloader includes a `DiskSpaceGuard` with backup/rollback protection and optional-package detection.

10 Verification Results

10.1 Summary

256 / 256 PASS	**0 FAIL**	**0 SKIP**
Seed: `1772952709`	Platform: g++ 14.2, WSL2 x86_64	

10.2 Suite G — FIRE Convergence Metrics

Table 3: FIRE relaxation of noble gas clusters from Fibonacci sphere starting geometry. All clusters converge to $F_{\text{rms}} < 10^{-5}$ kcal/mol/Å.

Cluster	U_0	U_f	F_{rms}	**Steps**	r_{nn} (Å)
Ar_2	−0.026	−0.238	9.98×10^{-6}	603	3.817
Ar_3	−0.696	−0.714	5.35×10^{-6}	308	3.817
Ar_4	−1.219	−1.428	4.78×10^{-6}	919	3.816
Ar_5	−1.675	−2.167	8.72×10^{-6}	1269	3.808
Ar_6	−2.077	−3.025	4.97×10^{-6}	2075	3.799
Ar_7	−2.462	−3.928	8.55×10^{-6}	2491	3.792

10.3 Suite M — NVE Energy Conservation

Table 4: NVE velocity-Verlet energy conservation on argon clusters. Maximum drift over 1000 steps at $\Delta t = 1$ fs.

System	E_0 (kcal/mol)	**Max drift**
Ar_3	1.167	2.91×10^{-12}
Ar_4	1.817	5.27×10^{-12}
Ar_5	0.314	2.26×10^{-11}

10.4 Suite I/J/K — Randomised Consistency

Table 5: Worst-case relative error across 500 randomised samples per suite.

Suite	**Property tested**	**Worst error**
I	LJ force–energy consistency	$< 10^{-3}$
J	Coulomb $U = kq_1q_2/r$	$< 10^{-6}$
K	Dielectric $U_{\text{vac}}/U_\varepsilon = \varepsilon_r$	$< 10^{-6}$

11 Demonstration Cases

11.1 Tier 1: Verification Anchor Systems

These systems are the primary validation targets, present in both the static empirical database and the PubChem/NIST downloaded cross-check.

Table 6: Tier 1 verification anchor molecules.

Molecule	**Bond**	r_{ref} **(Å)**	**Bonds**	**Source**
H_2	H–H	0.741	1	NIST
H_2O	O–H	0.958	2	PubChem
NH_3	N–H	1.012	3	PubChem
CH_4	C–H	1.087	4	PubChem
HF	H–F	0.917	1	NIST
HCl	H–Cl	1.275	1	NIST
CO_2	C=O	1.160	2	PubChem
NaCl	Na–Cl	2.361	–	NIST

11.2 Tier 2: Intermediate Molecular Demonstrations

These molecules exercise bond detection across functional groups, ring structures, and competing bond environments.

Table 7: Tier 2 molecular demonstrations (PubChem 3D cross-checked).

Molecule	**Bond tested**	r **(Å)**	**Feature**
Ethane (C_2H_6)	C–C	1.536	Single bond
Ethylene (C_2H_4)	C=C	1.339	Double bond
Acetylene (C_2H_2)	C≡C	1.203	Triple bond
Benzene (C_6H_6)	C–C	1.397	Aromatic ring
Methanol (CH_3OH)	C–O	1.428	Alcohol
Formaldehyde (CH_2O)	C=O	1.206	Carbonyl
Formic acid (HCOOH)	C–O	1.202	Carboxyl
Acetone	C=O	1.213	Ketone
HCN	C≡N	1.156	Nitrile
SO_2	S=O	1.432	Sulphur oxide
CS_2	C=S	1.554	Thiocarbonyl

11.3 Tier 3: Crystal Demonstrations

Table 8: Crystal preset verification. All atom counts and nearest-neighbour distances match Wyckoff references within 3%.

Crystal	**Type**	a **(Å)**	r_{nn} **(Å)**	**CN**	Z
Cu	FCC	3.615	2.556	12	4
Al	FCC	4.050	2.863	12	4
Fe	BCC	2.870	2.485	8	2
NaCl	Rocksalt	5.640	2.820	6	8
Si	Diamond	5.431	2.352	4	8
MgO	Rocksalt	4.212	2.106	6	8
CsCl	CsCl	4.123	3.571	8	2

11.4 Tier 4: Dielectric Screening Demonstrations

Table 9: Solvent dielectric screening verification. All ratios $U_{\text{vac}}/U_{\varepsilon} = \varepsilon_r$ to $< 10^{-6}$ relative error.

Solvent	ε_r	**Source**
Hexane	1.88	CRC 104
Benzene	2.27	CRC 104
Toluene	2.38	CRC 104
Diethyl ether	4.27	CRC 104
Chloroform	4.81	CRC 104
THF	7.58	CRC 104
Dichloromethane	8.93	CRC 104
Acetone	20.70	CRC 104
Ethanol	24.85	CRC 104
Methanol	32.66	CRC 104
Acetonitrile	35.69	CRC 104
DMF	36.70	CRC 104
Glycerol	46.53	CRC 104
DMSO	46.70	CRC 104
Water	78.40	CRC 104

12 Limitations

This section states the known boundaries of the current kernel.

1. **Metal/covalent crystal relaxation.** UFF LJ $r_{\min}$ exceeds the actual bond length for Cu ($r_{\min} = 3.92$ Å vs $r_{\text{nn}} = 2.56$ Å), Fe, and Si. FIRE relaxation drives these crystals away from physical equilibrium.

2. **No dipole interactions.** The current model treats atoms as point charges; induced dipoles and polarisation are not implemented.

3. **Incomplete charge equilibration.** QEq / Mulliken charge equilibration is present in the codebase but not yet exercised by the verification suite.

4. **No reactive chemistry.** Bond formation and breaking during dynamics is not supported; topology is fixed at initialisation.

5. **PBC not exercised in Suites G/N/Q.** Noble gas clusters and crystal builder tests run without periodic boundary conditions.

6. **Bonded force field coverage.** Harmonic bond/angle terms are present but not yet systematically verified against experimental equilibrium geometries.

7. **Suite G tests local descent only.** The Fibonacci-sphere starting geometry finds a local minimum, not necessarily the global LJ minimum for $N \geq 4$.

13 Future Work

1. **Bond/angle perturbation scans on polyatomics.** Extend Suite N to H_2O, NH_3 with bonded force fields.
2. **PBC crystal stability.** NVE dynamics in periodic supercells.
3. **Defect perturbation.** Vacancy and interstitial tests in NaCl and Cu supercells; verify topology preservation.
4. **QEq charge equilibration.** Verify convergence and Coulomb consistency under charge redistribution.
5. **Basin multiplicity.** Multi-start FIRE on Ar_7 to map the LJ local-minima landscape.
6. **Dipole and polarisation.** Point-dipole interactions and self-consistent field polarisation.
7. **Reactive MD.** Bond formation/breaking with energy-gated topology updates.
8. **Polymorph discovery.** Automated crystal structure search using formation priors and basin mapping.

14 Conclusion

The deterministic kernel of VSEPR-SIM has been verified for:

- Lennard-Jones interactions (42 UFF elements, force–energy consistency $< 10^{-3}$ relative error)
- Coulomb electrostatics (exact $1/r$ scaling, dielectric screening across 19 solvents)
- Crystal geometry (30 Wyckoff reference lattices, all within tolerance)
- Molecular bond detection (9 molecules, exact counts)
- FIRE relaxation (Ar_2–Ar_7, $F_{\mathrm{rms}} < 10^{-5}$, no deadlock)
- NVE energy conservation ($< 3 \times 10^{-11}$ kcal/mol drift)
- Empirical cross-check (25 PubChem bonds, 20 NIST diatomics)

All 485 verification checks pass without failure. The result is reproducible via seeded random suites and archived verbose output.

This establishes the first fully verified deterministic kernel baseline for VSEPR-SIM, completing the transition from foundational correctness recovery to a stable platform for convergence tightening, expanded physical coverage, and formation-oriented workflow integration.

Reproduce this result:

```
wsl bash scripts/run_all_verification.sh --full
wsl bash -c "cd␣wsl-build␣&&␣./deep_verification␣--seed␣1772952709␣--rand-iters␣500"
```

A UFF Lennard-Jones Parameters (42 Elements)

Element	σ (Å)	ε (kcal/mol)	$r_{\min}$ (Å)
H	2.886	0.044	3.239
He	2.362	0.056	2.651
Li	2.451	0.025	2.751
Be	2.745	0.085	3.081
B	3.637	0.180	4.082
C	3.431	0.105	3.851
N	3.261	0.069	3.660
O	3.118	0.060	3.500
F	2.997	0.050	3.364
Ne	3.243	0.042	3.640
Na	3.328	0.030	3.735
Mg	3.021	0.111	3.391
Al	4.499	0.505	5.049
Si	3.826	0.402	4.295
P	3.695	0.305	4.147
S	3.595	0.274	4.035
Cl	3.516	0.227	3.947
Ar	3.400	0.238	3.817
K	4.241	0.035	4.760
Ca	3.399	0.238	3.816
Ti	2.829	0.017	3.175
V	2.801	0.016	3.143
Cr	2.693	0.015	3.022
Mn	2.638	0.013	2.960
Fe	2.594	0.013	2.912
Co	2.559	0.014	2.872
Ni	2.525	0.015	2.834
Cu	3.114	0.005	3.495
Zn	2.462	0.124	2.763
Ga	3.905	0.415	4.384
Ge	3.813	0.379	4.280
As	3.769	0.309	4.230
Se	3.746	0.291	4.204
Br	3.732	0.251	4.189
Kr	3.689	0.220	4.141
Rb	4.114	0.040	4.617
Sr	3.641	0.235	4.088
Mo	2.719	0.056	3.052
Pd	2.583	0.048	2.899
Ag	2.805	0.036	3.149
I	4.009	0.339	4.500
Xe	3.924	0.332	4.405

Source: Rappé et al., *J. Am. Chem. Soc.* **114**, 10024 (1992). All values verified by Suite A.

B Empirical Bond Length Database

57 reference bond lengths from NIST CCCBDB, Wyckoff crystal data, and PubChem 3D conformers. See `atomistic/core/empirical_reference.hpp` for the complete table with source annotations and tolerance bounds.

C CLI and Verification Usage

```
# Quick verification (no network)
wsl bash scripts/run_all_verification.sh

# Full verification with PubChem/NIST download
wsl bash scripts/run_all_verification.sh --full

# Deep verification with specific seed
./wsl-build/deep_verification --seed 1772952709 --verbose

# Fetch empirical references
python3 scripts/fetch_empirical.py --offline  # fallback only
python3 scripts/fetch_empirical.py            # live PubChem + NIST
python3 scripts/fetch_empirical.py --verbose  # show each fetch
```

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